SWSS ORCHAGENT AND SYNCD EXTENSION PROPOSAL FOR OPTICAL CIRCUIT SWITCH

SONiC-OCS Sub-group Meeting

September 22nd, 2026

Coherent Team

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OVERALL INFORMATION

- **SWSS Orchagent Extension for OCS**
- **SAI/Syncd Extension for OCS**

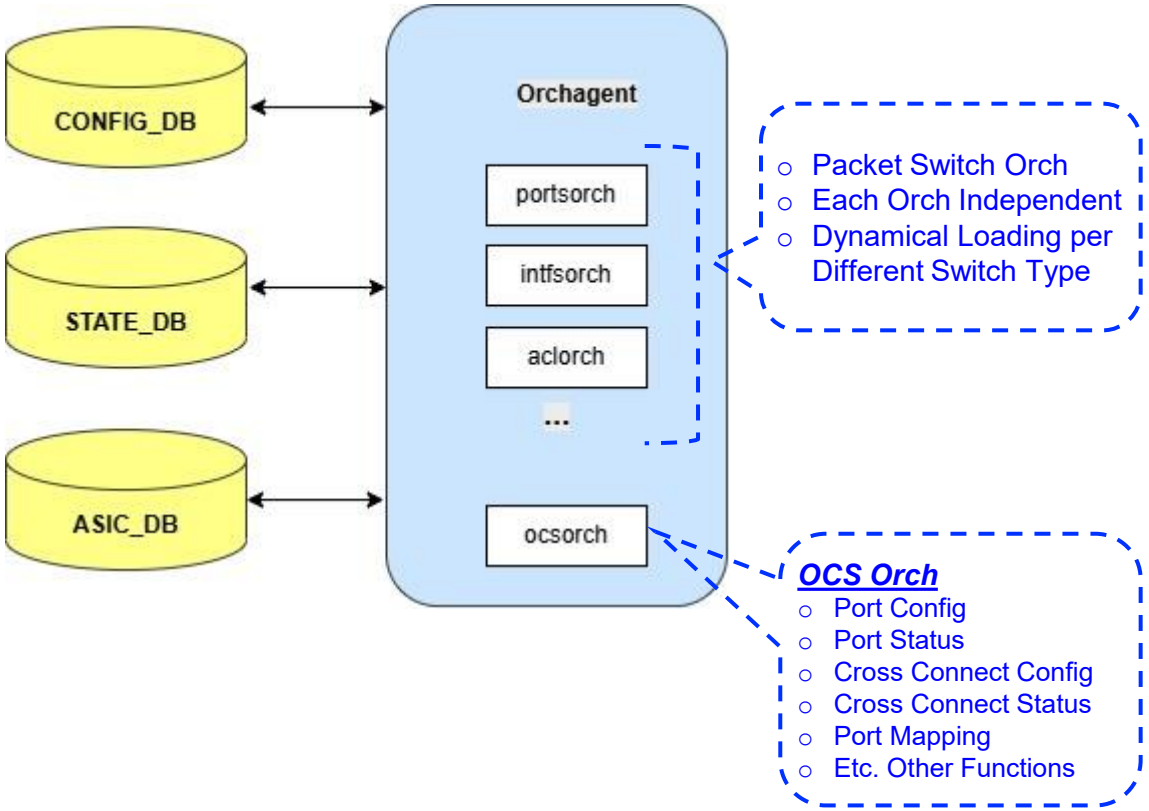
More Information and Materials

- **HLD for OCS**
  - OCS High Level Design https://github.com/sonic-net/SONiC/pull/2269
- **SONiC-OCS Developer Guide**
  - https://github.com/sonic-ocs/SONiC/blob/ocs-dev/doc/ocs/sonic-ocs-developer-guide.md
- **SAI API for OCS**
  - https://github.com/opencomputeproject/SAI/tree/master/doc/OCS
  - https://github.com/opencomputeproject/SAI/commit/4074f6d22dee267a6099222acb03200e11f4c72e
- **On-going PRs for Upstream and SONiC-OCS Dev**
  - https://github.com/sonic-net/sonic-buildimage/pull/29514
  - https://github.com/sonic-ocs/sonic-buildimage/pull/3
  - https://github.com/sonic-ocs/sonic-swss-common/pull/1

SWSS ORCHAGENT EXTENSION FOR OCS – 1/2

- **Follow SWSS Orchagent current design / implementation with extension for OCS.**
- **OCS Orch and Packet Switch Orch independent.**
- DEVICE_METADATA **switch_type defined for different switches to dynamically load the orchestration modules for different functions.**
- ***Question:*** **new switch-type for OCS device in sonic-device_metadata.yang to be added. Or other way in another attribute?**

```
leaf switch_type {
    type string {
        pattern "chassis-packet|fabric|npu|voq|dpu|dummy-sup";
    }
    description "Type of switch. Default is NPU, on a VOQ switch voq is used for a regular
                 switching device while fabric is used for a fabric device.
                 chassis-packet is used for chassis in packet mode.";
}
```

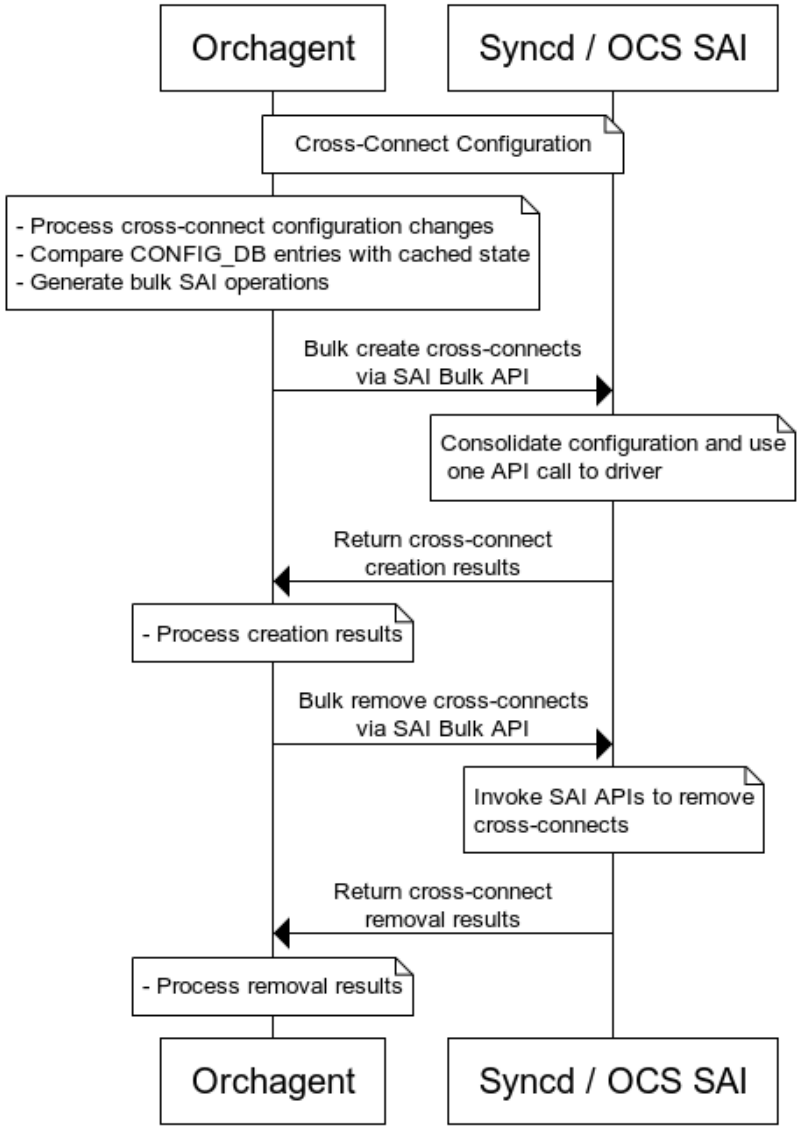


SWSS ORCHAGENT EXTENSION FOR OCS – 2/2

- **OCS Orchagent Consolidated Configuration Processing with Multiple Notifications**
  - ❑ Redis keyspace notification and callback per DB entry based.
  - ❑ In each callback, reading all CONFIG_DB cross-connect entries, compares with cache to find the changes
  - ❑ If there're changes, continue further processing to apply the changes, then update cache
  - ❑ If no changes are found, no more processing
- ***Questions:***
  - ❑ Need gNMI batch commits OCS cross-connect to CONFIG_DB through Redis transaction. Then, batch commit need be distinguished for different specific scenarios or tables as framework level enhancement.
  - ❑ If a batch commit is there, how about solution to do enhancement on sonic-swss-common for one time notification instead of per DB entry?

SAI/SYNCD EXTENSION FOR OCS – BATCH CROSS CONNECT CONFIGURATION

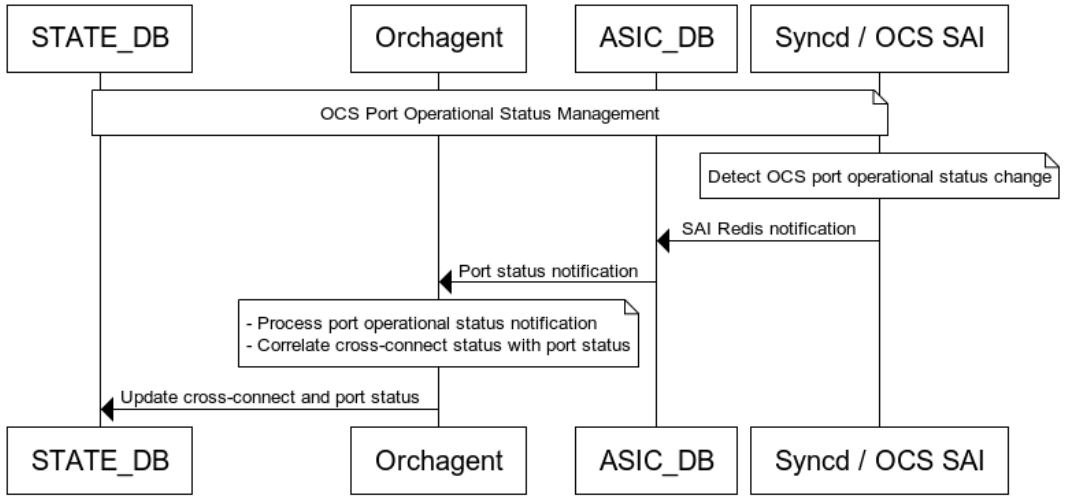
- **SAI/Syncd One-time Batch Configuration:**
  - **DB Change Processing:** During callback handling, it compares cached entries against new CONFIG_DB entries then invokes bulk creation APIs for all CONFIG_DB cross-connect entries (via SAI error mode: SAI_BULK_OP_ERROR_MODE_IGNORE_ERROR). After this bulk remove is needed to delete SAI related objects from logical view, instead of operations to module or hardware.
  - **SAI Bulk APIs Applying:** Based on this sequence, the vendor SAI library can consolidate configuration into a single call to apply updates to the OCS module as one-time access for better performance.
- ***Question:*** **Potential enhancement on SAI Bulk APIs to support mixed scenarios, like mixed "create", "set", "remove" operations in one bulk API.**



SAI/SYNCD EXTENSION FOR OCS – NOTIFICATION

- **Follow SONiC notification mechanism for OCS**
  - ❑ Syncd registers the notification callback to vendor driver for function monitoring
  - ❑ Vendor driver monitoring status. When it is changed, driver triggers the notification callback.
  - ❑ Syncd sending the notification through sairedis framework to OCS Orchagent.
  - ❑ Orchagent processing the notification and updating the status change to STATE_DB.

  OCS port status as one example for this solution
- **More new functions for monitoring can be further discussed as next steps.**
  - ❑ Notification, like alarm, event, status etc.
  - ❑ Syncd Polling, like counter or performance etc.




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